

This document is a **transcript** of the lecture, with extra summary and vocabulary sections for your convenience. Due to time constraints, the notes may sometimes only contain limited illustrations, proofs, and examples; for a more thorough discussion of the course content, **consult the textbook**.

Summary

Quick summary of today's notes. Lecture starts on next page.

- The *inner product* of two vectors $u, v \in \mathbb{R}^n$ is $u \bullet v = u_1v_1 + u_2v_2 + \cdots + u_nv_n \in \mathbb{R}$.

A set of nonzero vectors $v_1, v_2, \dots, v_p \in \mathbb{R}^n$ is *orthogonal* if $v_i \bullet v_j = 0$ for all $i \neq j$.

An orthogonal basis is *orthonormal* if it consists entirely of unit vectors.

If $x_1, x_2, \dots, x_p$ is a basis for a subspace $V \subseteq \mathbb{R}^n$ then the vectors

$$\begin{aligned} v_1 &= x_1, \\ v_2 &= x_2 - \frac{x_2 \bullet v_1}{v_1 \bullet v_1} v_1, \\ v_3 &= x_3 - \frac{x_3 \bullet v_1}{v_1 \bullet v_1} v_1 - \frac{x_3 \bullet v_2}{v_2 \bullet v_2} v_2, \\ &\vdots \\ v_p &= x_p - \frac{x_p \bullet v_1}{v_1 \bullet v_1} v_1 - \frac{x_p \bullet v_2}{v_2 \bullet v_2} v_2 - \cdots - \frac{x_p \bullet v_{p-1}}{v_{p-1} \bullet v_{p-1}} v_{p-1}, \end{aligned}$$

form an orthogonal basis for V .

- Suppose A is an $m \times n$ matrix and $b \in \mathbb{R}^n$.

It may happen that the linear system $Ax = b$ has no solutions $x \in \mathbb{R}^n$.

When this occurs, one wants to find an approximate solution.

Define the *length* of $v \in \mathbb{R}^n$ by $\|v\| = \sqrt{v \bullet v}$.

A good approximate solution to $Ax = b$ is a vector $s \in \mathbb{R}^n$ that makes $\|As - b\|$ as small as possible.

We call such a vector $s \in \mathbb{R}^n$ a *least-squares solution* to $Ax = b$.

- **The least-squares solutions to $Ax = b$ are just the ordinary solutions to $A^\top Ax = A^\top b$.**

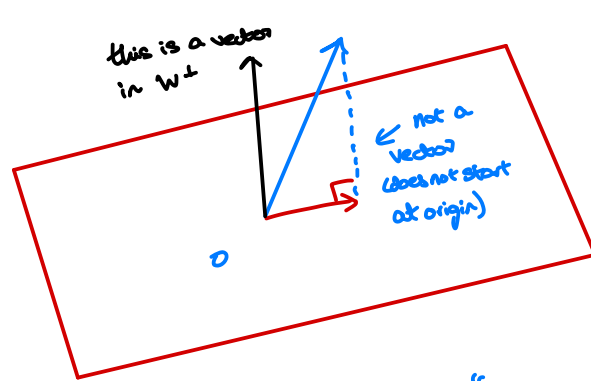
So to find least-squares solutions, just form the matrix $\begin{bmatrix} A^\top A & A^\top b \end{bmatrix}$ and row reduce as usual.

- If $Ax = b$ is a consistent linear system then it has the same solutions as $A^\top Ax = A^\top b$.

So if $Ax = b$ is consistent then least-squares solutions are the same as ordinary/exact solutions.

- However, the system $A^\top Ax = A^\top b$ is always consistent even if $Ax = b$ is not.

So there is **always** at least one least-squares solution, even if $Ax = b$ has no exact solutions.



is only vector in $W^\perp$ you can
 get by adding elements of
 W to y

black vector = $y - \text{proj}_W(y) \in W^\perp$

$\text{proj}_W(y)$ = "shadow of y cast on W "

Orthogonal: $v_i \cdot v_j = 0$ if $i \neq j$
 orthonormal: $v_i \cdot v_j = 0$ if $i \neq j$ AND $v_i \cdot v_i = 1$ if $i = j$

1 Last time: orthonormal vectors, projections, orthogonal bases

Vectors $u_1, u_2, \dots, u_p$ are *orthonormal* if each vector is a unit vector and any two vectors are orthogonal.

In other words, if $u_i \cdot u_j = 0$ when $i \neq j$ and $u_i \cdot u_i = 1$ for all $i = 1, 2, \dots, p$.

The standard basis $e_1, e_2, \dots, e_n$ of $\mathbb{R}^n$ consists of orthonormal vectors.

If $v_1, v_2, \dots, v_p$ are orthogonal and all nonzero then $\frac{1}{\|v_1\|}v_1, \frac{1}{\|v_2\|}v_2, \dots, \frac{1}{\|v_p\|}v_p$ are orthonormal.

Theorem. Let U be an $m \times n$ matrix. The columns of U are orthonormal vectors if and only if $U^T U = I_n$.

If this happens then $(Ux) \cdot (Uy) = x \cdot y$ for all $x, y \in \mathbb{R}^n$. $\rightarrow U$ preserves lengths and angles (and inner product) $U^{-1} = U^T$

If $m = n$ then $U^T U = I_n$ means $U^{-1} = U^T$, in which case we say that U is an orthogonal matrix.

Example. The columns of $U = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$ are orthonormal as $U^T U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. $\rightarrow$ Square + orthonormal columns.

$\rightarrow$ orthogonal complement.

Let $W \subseteq \mathbb{R}^n$ be any subspace. Recall that $W^\perp = \{w \in \mathbb{R}^n : w \cdot v = 0 \text{ for all } v \in W\}$.

We showed last time that $W \cap W^\perp = \{0\}$ and $\dim W + \dim W^\perp = n$. $\rightarrow$ dimensions sum to n .

$\rightarrow$ disjoint subspaces

Theorem. Let $y \in \mathbb{R}^n$. Then there is a unique vector $\text{proj}_W(y) \in W$ such that $y - \text{proj}_W(y) \in W^\perp$.

We call $\text{proj}_W(y)$ the *orthogonal projection* of y onto W .

To compute $\text{proj}_W(y)$, use this fact: if $u_1, u_2, \dots, u_p$ is any orthogonal basis of W then

$\text{proj}_W(y) = y \iff y \in W$

$$\text{proj}_W(y) = \frac{y \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{y \cdot u_2}{u_2 \cdot u_2} u_2 + \dots + \frac{y \cdot u_p}{u_p \cdot u_p} u_p.$$

algorithmic definition.

This formula works for any choice of orthogonal basis for W , but it requires you to find such a basis.

optimization definition of $\text{proj}_W(y)$

Properties of orthogonal projections

$\text{proj}_W(y)$ is the unique solution to $\min \|y - \vec{w}\|$ subject to $\vec{w} \in W$.

We have $\text{proj}_W(y) = y$ if and only if $y \in W$, and $\text{proj}_W(y) = 0$ if and only if $y \in W^\perp$.

Recall $\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$. Then $\|y - \text{proj}_W(y)\| < \|y - v\|$ for all $v \in W$ with $v \neq \text{proj}_W(y)$.

To compute $\text{proj}_W(y)$, need an orthogonal basis for W .

Gram-Schmidt process.

Let $W \subseteq \mathbb{R}^n$ be a subspace. Suppose $x_1, x_2, \dots, x_p$ is a basis for W .

Input: Any basis
 Output: Orthogonal basis.

Define $v_1, v_2, \dots, v_p \in W$ inductively by the following formulas:

$$v_1 = x_1.$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1.$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2.$$

$\vdots$

$$v_p = x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} \overset{v_1}{\uparrow} - \frac{x_p \cdot v_2}{v_2 \cdot v_2} \overset{v_2}{\uparrow} - \frac{x_p \cdot v_3}{v_3 \cdot v_3} \overset{v_3}{\uparrow} - \dots - \frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} v_{p-1}.$$

For each $i = 1, 2, \dots, p$, the vectors $v_1, v_2, \dots, v_i$ are an orthogonal basis for the subspace

$$\mathbb{R}\text{-span}\{x_1, x_2, \dots, x_i\} = \mathbb{R}\text{-span}\{v_1, v_2, \dots, v_i\}.$$

Consequently v_{i+1} is just x_{i+1} minus the orthogonal projection of x_{i+1} onto this subspace.

The full list of vectors $v_1, v_2, \dots, v_p$ is an orthogonal basis for W .

$W = [\text{vectors in } \mathbb{R}^4 \text{ where entries sum to } 0]$.

Example. Let $W = \text{Nul} \left(\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \right) = \{w \in \mathbb{R}^4 : w_1 + w_2 + w_3 + w_4 = 0\}$.

A basis for W is given by $x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$, $x_2 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$.

To find an orthogonal basis, we let

$$v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}.$$

$$v_2 = x_2 - \frac{x_2 \bullet v_1}{v_1 \bullet v_1} v_1 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \frac{0 - 1 + 0 + 0}{1 + 1 + 0 + 0} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix}.$$

$$v_3 = x_3 - \underbrace{\frac{x_3 \bullet v_1}{v_1 \bullet v_1}}_{=0} v_1 - \frac{x_3 \bullet v_2}{v_2 \bullet v_2} v_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix}.$$

Thus $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix}$ are an orthogonal basis for W .

The rescaled vectors $\begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \\ -2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 1 \\ 1 \\ -3 \end{bmatrix}$ are also an orthogonal basis for W .

*Check: ① Vectors are in W
② Orthogonal.*

for orthonormal divide by distance of each vector.

2 Least-squares problems

more equations than variables.

Many linear systems $Ax = b$ that arise in applications are *overdetermined* (meaning they have more equations than variables, so the matrix A has more rows than columns) and often inconsistent (meaning they have no exact solution $x \in \mathbb{R}^n$).

For example, $b \in \mathbb{R}^m$ might be a vector of measurements and each row of Ax might provide an approximation for what we expect these measurements to be in terms of certain inputs $x \in \mathbb{R}^n$.

Because measurements are noisy and because our linear approximations are inexact, there may be no input vector $x \in \mathbb{R}^n$ such that $Ax = b$. When no exact solution is available, the next best thing to provide is an input vector $x \in \mathbb{R}^n$ such that Ax is as “close” to the vector $b \in \mathbb{R}^m$ as possible.

There is more than one way to quantify how close two vectors are to each other. One common method is to use the distance function we have already seen: define the *distance* between vectors $u, v \in \mathbb{R}^n$ to be

$$\|u - v\| = \sqrt{(u - v) \bullet (u - v)} = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2}.$$

Two vectors are close if their distance in this sense is small. The distance function $\|\cdot\|$ is called the *Euclidean distance* or *L^2 -distance*. In two and three dimensions, this distance is the usual way that we measure distance between points in space.

Vector minimising $\text{dist}(Au, b) = \|b - Au\| = \|Au - b\|$.

Definition. If A is an $m \times n$ matrix and $b \in \mathbb{R}^m$, then a *least-squares solution* to the linear system $Ax = b$ is a vector $s \in \mathbb{R}^n$ such that $\|b - As\| \leq \|b - Ax\|$ for all $x \in \mathbb{R}^n$.

In other words, a least-squares solution to $Ax = b$ is a vector $s \in \mathbb{R}^n$ that minimizes $\|b - As\|$.

A vector that minimizes $\|b - As\|$ will also minimize $\|b - As\|^2$, which is the sum of the squares of the entries in the vector $b - As$. This accounts for the name “least-squares.”

Least-squares problems (that is, problems requiring us to find a least-squares solution to some linear system) arise frequently in engineering and statistics. Being able to solve such problems is an important application of the material covered in this course. Our goal today is to describe the general solution to the least-squares problem. Here are the key points:

- If $Ax = b$ is a consistent linear system then every least-squares solution will be an exact solution.
- There may be more than one least-squares solution to a given linear system $Ax = b$.
- However, in contrast to exact solutions, there is **always** at least one least-squares solution.

The last fact is not obvious from the definition of a least-squares solution.

Solving least-squares problems in general.

Fix an $m \times n$ matrix A and a vector $b \in \mathbb{R}^m$

A least-squares solution $s \in \mathbb{R}^n$ to $Ax = b$ is a vector such that $\|As - b\|$ is as small as possible.

If $s \in \mathbb{R}^n$ then we necessarily have $As \in \text{Col } A$.

As mentioned earlier, if $v \in \text{Col } A$ minimizes $\|v - b\|$ then we must have $v = \text{proj}_{\text{Col } A}(b)$. Therefore:

Lemma. The least-squares solutions to $Ax = b$ are precisely those $s \in \mathbb{R}^n$ such that $As = \text{proj}_{\text{Col } A}(b)$.

Using this lemma, we can prove something even more explicit:

Theorem. The set of least-squares solutions to $Ax = b$ is the set of exact solutions to the linear system $A^T Ax = A^T b$. This new linear system is always consistent so its set of solutions is nonempty.

Proof. Let $\hat{b} = \text{proj}_{\text{Col } A}(b)$. Since $b - \hat{b} \in (\text{Col } A)^\perp = \text{Nul } A^T$, we have $A^T(b - \hat{b}) = 0$ and $A^T \hat{b} = A^T b$.

Thus, if $s \in \mathbb{R}^n$ satisfies $As = \hat{b}$ then $A^T As = A^T \hat{b} = A^T b$.

Conversely, if $s \in \mathbb{R}^n$ satisfies $A^T As = A^T b$, then $A^T(As - b) = 0$ so $As - b \in \text{Nul } A^T = (\text{Col } A)^\perp$.

In this case, it follows by the uniqueness of orthogonal projections that $As = \text{proj}_{\text{Col } A}(b) = \hat{b}$.

This shows that the set of exact solutions to $Ax = \hat{b}$, which is precisely the set of least-squares solutions to $Ax = b$, is the same as the set of exact solutions to $A^T Ax = A^T b$.

We claimed that the linear system $A^\top Ax = A^\top b$ is always consistent. This holds since $\widehat{b} \in \text{Col } A$, so by definition there must exist some $s \in \mathbb{R}^n$ such that $As = \widehat{b}$, and it then holds that $A^\top As = A^\top \widehat{b} = A^\top b$. $\square$

Example. Here is a simple, somewhat contrived example.

$$\text{Let } A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \text{ and } b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}.$$

To find a least-squares solution to $Ax = b$, we compute

$$A^\top A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \quad \text{and} \quad A^\top b = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}.$$

The least-squared solutions we want are the exact solutions to $A^\top Ax = A^\top b$. Solve this by row reducing:

$$\begin{bmatrix} 17 & 1 & 19 \\ 1 & 5 & 11 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 11 \\ 17 & 1 & 19 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 11 \\ 0 & -84 & -168 \end{bmatrix} \sim \begin{bmatrix} 1 & 5 & 11 \\ 0 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

In this case $A^\top Ax = A^\top b$ has a unique solution

$$s = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

which is also the unique least-squares solution to $Ax = b$. Note that $As \neq b$ as

$$\|As - b\| = \left\| \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} \right\| = \left\| \begin{bmatrix} 2 \\ 4 \\ -8 \end{bmatrix} \right\| = \sqrt{4 + 16 + 64} = \sqrt{84}.$$

Geometrically, we interpret the least-squares solution as meaning that

$$As = \begin{bmatrix} 4 \\ 4 \\ 3 \end{bmatrix}$$

is the point in the plane spanned by the columns of A in $\mathbb{R}^3$ that is closest to b .

A linear system $Ax = b$ has a unique solution for every $b \in \mathbb{R}^m$ if and only if the matrix A is invertible. The following theorem describes, analogously, when $Ax = b$ has a unique least-squares solution.

Theorem. Let A be an $m \times n$ matrix. The following are then equivalent:

- (a) $Ax = b$ has a unique least-squares solution for each $b \in \mathbb{R}^m$.
- (b) The columns of A are linearly independent.
- (c) $A^\top A$ is invertible.

When these properties hold, the unique least-squares solution to $Ax = b$ is $s = (A^\top A)^{-1} A^\top b$, which is the unique exact solution to $A^\top Ax = A^\top b$.

Remark. In practice, the product $(A^\top A)^{-1} A^\top b$ is never computed directly for a large linear system. It is more efficient to find s by solving the system $A^\top Ax = A^\top b$ via row reduction.

Proof. Suppose $s \in \mathbb{R}^n$ is a least-squares solution to $Ax = b$.

If $v \in \text{Nul } A$ then $s + v$ is also a least-squares solution since $\|As - b\| = \|A(s + v) - b\|$.

Therefore if (a) holds then we must have $\text{Nul } A = \{0\}$ so (b) must also hold.

If $v \in \mathbb{R}^n$ then $A^\top Av = 0$ if and only if $Av \in \text{Col } A \cap \text{Nul } A^\top = \text{Col } A \cap (\text{Col } A)^\perp = \{0\}$.

Therefore $\text{Nul}(A^\top A) = \text{Nul}(A)$. Hence if (b) holds then $\text{Nul}(A^\top A) = \text{Nul } A = \{0\}$.

In this case, $A^\top A$ is invertible since it is a square matrix, so if (b) holds then (c) holds.

Finally, if (c) holds then the linear system $A^\top Ax = A^\top b$ has a unique solution so (a) holds by the previous theorem. The chain of implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (a) shows that the properties are equivalent. $\square$

It is often easier to compute a least-squares solution to $Ax = b$ if the columns of A are orthogonal:

Example. Suppose $A = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}$ and $b = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$. The columns of A are orthogonal.

The orthogonal projection of b onto $\text{Col } A$ is therefore

$$\text{proj}_{\text{Col } A}(b) = \frac{\begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{\begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} -6 \\ -2 \\ 1 \\ 7 \end{bmatrix}}{\begin{bmatrix} -6 \\ -2 \\ 1 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -6 \\ -2 \\ 1 \\ 7 \end{bmatrix}} \begin{bmatrix} -6 \\ -2 \\ 1 \\ 7 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} -6 \\ -2 \\ 1 \\ 7 \end{bmatrix}.$$

We deduce that $s = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$ is a least-squares solution to $Ax = b$ since $As = \text{proj}_{\text{Col } A}(b)$.

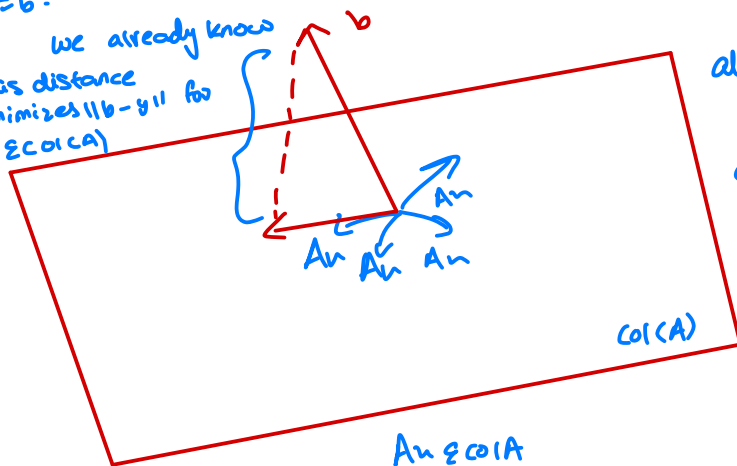
why do least squares solution exist?

Solve $Au = b$.

minimizing $\|b - y\|$ for $y \in \text{col}(A)$

$\Leftrightarrow \min \|b - Au\|$ for $u \in \mathbb{R}^n$

we already know
this distance
minimizes $\|b - y\|$ for
 $y \in \text{col}(A)$



also. $\text{proj}_{\text{col}(A)} b \in \text{col}(A)$, so exists
 u with $Au = \text{proj}_{\text{col}(A)} b$ and
every such u is a least
square solution.

3 Vocabulary

Keywords from today's lecture:

1. **Overdetermined** linear system.

A linear system with more equations than variables.

2. **Least-squares solution** to a linear system $Ax = b$.

Assume A is an $m \times n$ matrix and $b \in \mathbb{R}^m$. Two equivalent definitions:

- A vector $s \in \mathbb{R}^n$ is a least-squares solution to $Ax = b$ if $\|b - As\| \leq \|b - Ax\|$ for all $x \in \mathbb{R}^n$.
- A vector $s \in \mathbb{R}^n$ is a least-squares solution to $Ax = b$ if s is an exact solution to the (always consistent) linear system $A^\top Ax = A^\top b$.

To find a least-squares solution to $Ax = b$, use row reduction to find a solution to $A^\top Ax = A^\top b$